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Section _____

MTH:2310 DISCRETE MATH
QUIZ 9
DUE: FRIDAY NOV. 21

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. (6 points) Binomial Coefficients and Identities

What is the coefficient of z^5x^{12} in the expansion of $(3z - 2x^3)^9$?

$$\sum_{k=0}^9 \binom{9}{k} (3z)^{9-k} (-2x^3)^k$$

$$\binom{9}{4} (3z)^5 (-2x^3)^4$$

we get the z^5x^{12} term
 when $k=4$

thus the coefficient is

$$\binom{9}{4} (3)^5 (-2)^4$$

Problem 2. (8 points) *Binomial Coefficients and Identities*

Evaluate the sum.

$$17^5 \cdot 5^{17} \cdot \left(\binom{17}{0} - \frac{4}{5} \binom{17}{1} + \frac{16}{25} \binom{17}{2} - \frac{64}{125} \binom{17}{3} + \cdots - \frac{4^{17}}{5^{17}} \binom{17}{17} \right)$$

$$\sum_{k=0}^{17} \binom{17}{k} X^{17-k} Y^k \quad Y = -\frac{4}{5} \cdot 17^{-12} \cdot 5 \quad X = 17^{-12} \cdot 5$$

$$\left(\frac{5}{17^{12}} - \frac{4}{17^{12}} \right) = \left(\frac{1}{17^{12}} \right) = 17^5$$

Problem 3. (6 points) *Permutations and Combinations*How many different four-digit PIN's can be made using distinct digits 0 - 9? Express your answer using the notation $P(n, r)$ or $C(n, r)$ or $\binom{n}{r}$. Clearly justify your answer.

in the case of a PIN,
order matters, thus we use
permutations

$$P(10, 4) = \frac{10!}{4!}$$

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Problem 4. (6 points) *Permutations and Combinations*

How many different values can be made by multiplying exactly three distinct numbers from 2, 3, 5, 7, and 11? Express your answer using the notation $P(n, r)$ or $C(n, r)$ or $\binom{n}{r}$. Clearly justify your answer.

Multiplication for order doesn't matter,
therefore we use a combination, NOT
a permutation.

$$|\{2, 3, 5, 7, 11\}| = 5$$

$$\binom{5}{3} = \frac{5!}{3!(5-3)!}$$

Problem 5. (16 points) *Permutations and Combinations*

- (a) (8 points) How many 10-digit positive integers are there with exactly 4 zeros, if every other digit may only appear once? Express your answer using the notation $P(n, r)$ or $C(n, r)$ or $\binom{n}{r}$. Clearly justify your answer.

Choose position of the zeros

$$\binom{10}{4}$$

Choose the digits to use

$$\binom{9}{6}$$

Choose positions for the digits
6!

If the first digit is zero, we have

$$\binom{9}{3} \binom{9}{6} (6!) \text{ possibilities}$$

thus the number of permutations is the total permutations minus 24 permutations with a leading zero.

$$\binom{10}{4} \binom{9}{6} (6!) - \binom{9}{3} \binom{9}{6} (6!)$$

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- (b) (8 points) How many of the integers from part (a) are **not** palindromes? Express your answer using the notation $P(n, r)$ or $C(n, r)$ or $\binom{n}{r}$. Clearly justify your answer.

for the integer to be a palindrome,
the latter 5 digits are dependent on the former 5,
there we could find the permutations for
the first five

$$\binom{5}{2} \binom{9}{3} 3!$$

No leading zero:

$$\binom{4}{2} \binom{9}{3} 3!$$

$$\# \text{ palindromes} = \binom{5}{2} \binom{9}{3} 3! - \binom{4}{2} \binom{9}{3} 3!$$

